

Tempo Driven Audio-to-Score Alignment Using Spectral Decomposition and Online Dynamic Time Warping

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In this article, we present an online score following framework designed to deal with automatic accompaniment. The proposed framework is based on spectral factorization and online Dynamic Time Warping (DTW) and has two separated stages: preprocessing and alignment. In the first one, we convert the score into a reference audio signal using a MIDI synthesizer software and we analyze the provided information in order to obtain the spectral patterns (i.e., basis functions) associated to each score unit. In this work, a score unit represents the occurrence of concurrent or isolated notes in the score. These spectral patterns are learned from the synthetic MIDI signal using a method based on Non-negative Matrix Factorization (NMF) with Beta-divergence, where the gains are initialized as the ground-truth transcription inferred from the MIDI. On the second stage, a non-iterative signal decomposition method with fixed spectral patterns per score unit is used over the magnitude spectrogram of the input signal resulting in a distortion matrix that can be interpreted as the cost of the matching for each score unit at each frame. Finally, the relation between the performance and the musical score times is obtained using a strategy based on online DTW, where the optimal path is biased by the speed of interpretation. Our system has been evaluated and compared to other systems, yielding reliable results and performance.

CCS Concepts: • **Applied computing** → Sound and music computing

Additional Key Words and Phrases: Accompaniment, audio-to-score alignment, score following, online algorithm, non-negative matrix factorization (NMF), beta-divergence, dynamic time warping (DTW), speed of interpretation, tempo

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1. INTRODUCTION

Interactive Music Systems encompass the use of realtime technology in music composition and live performance. Often, this “interactivity” is associated with an electronic equipment driven by realtime software “listening” and responding to the performer’s input. In the particular case of score-driven interaction, the core of this “intelligence” is the score following. In this case, the electronics are defined as events that are triggered at predefined moments according to some sort of virtual time line. For example, in the case of automatic accompaniment, these events are no more than notes to be sent to a synthesizer or sampler, while for realtime tracking, they are just the current position of the performance in the score sheet.

In this work, we address the problem of audio-to-score alignment (or score matching), which is the task of synchronizing an audio recording of a musical piece with the corresponding symbolic score. There are two approaches to this problem, often called “offline” and “online” alignment. In offline alignment, the whole performance is accessible for the alignment process, i.e., it allows us to “look into the future” while establishing the matching. Therefore, this allows the development of non-causal algorithms that can reach higher matching precision [Joder et al. 2011]. This is interesting for applications that do not require the realtime property, such as Query-by-Humming [McNab et al. 1996], intelligent audio editors [Dannenberg 2007], and as a front-end for many Music Information Retrieval (MIR) systems. Online alignment, also known as score following, processes the data as the signal is acquired. This tracking is very useful for applications such as automatic page turning [Arzt 2008] or automated computer accompaniment of a live soloist [Raphael 2010]. Other possible applications include synchronization of live sound processing algorithms for instrumental electroacoustic composition or the control of visual effects synchronized with the music (e.g., stage lights or opera supertitles).

Audio-to-score alignment is traditionally performed in two steps: feature extraction and alignment. On the one hand, the features extracted from the audio signal characterize some specific information about the musical content. Different representations of the audio frame have been used, such as the output of a Short-Time Fourier Transform (STFT) [Cont 2010], auditory filter bank responses [Montecchio and Orio 2009], chroma vectors [Hu et al. 2009; Izmirli and Dannenberg 2010], or multi-pitch analysis information [Puckette 1995; Duan and Pardo 2011]. On the other hand, the alignment is performed by finding the best match between the feature sequence and the score. In fact, classical offline systems rely on cost measures between events in the score and in the performance. Two methods well known in speech recognition have been extensively used in the literature: statistical approaches (e.g., Hidden Markov Models (HMMs)) [Cont 2006; Duan and Pardo 2011; Joder et al. 2013; Cuvillier and Cont 2014] and Dynamic Time Warping (DTW) [Hu et al. 2009; Orio and Schwarz 2001; Dixon 2005; Carabias-Orti et al. 2015]. Although these techniques achieve good results, their best performances are obtained in offline applications; the results decrease significantly for real-time audio-to-score alignment.

In this work, we use the term “online” referring to those approaches that perform the estimation from only the current and previous frames of audio. Note that an online system can be run in realtime with the appropriated implementation and hardware. The prediction time is here defined as the difference between the onset time assigned to a note event by the alignment system and the time it reports about this note event. In other words, the estimated onset times given by an alignment system may not coincide with the actual occurrence of the note onsets in the audio. The prediction time could be a highly important factor for several applications. For example, in the case of page turning, a negative prediction time of about one hundred milliseconds is acceptable

[Arzt 2008]. However, for automatic accompaniment, the delay of the decision is critical [Cont 2008], and the system should be designed for positive prediction times. In this article, we extend the work presented in Carabias-Orti et al. [2015] by introducing constraints related to the speed of interpretation, which can be seen as a proxy for tempo changes. As a result, we present a novel alignment framework that allows different prediction times to deal with the aforementioned applications.

In particular, the proposed online score following framework is based on two stages: preprocessing and realtime alignment. On the first stage, we analyze the provided MIDI score and extract the sequence of score units (also referred to in this work as units), that is, the sequence of isolated notes and combinations of concurrent notes implicitly listed in the MIDI score. Then, the score is converted into a reference audio signal using a synthesizer software and we use a method based on Non-negative Matrix Factorization (NMF) with Beta-divergence to learn spectral patterns (i.e., basis functions) for each state. A similar approach was used by Fritsch and Plumbley [2013], but they use one component per instrument and note, plus some extra components to model the residual sounds. Note that the association of components to units rather than notes is of high importance here because it lets us restrict the model to have only one component active at a time (single, non-zero restriction), and then decomposition can be performed in a non-iterative manner as proposed in Carabias-Orti et al. [2013]. This is more efficient than iterative algorithms. NMF was also used by Cont [2006] as a multi-pitch estimator which defines the observation model. Joder et al. [2013] also defined a set of template vectors for each combination of concurrent notes, but directly from the score (i.e., without using audio synthesis). The unit templates are obtained as a linear mapping of individual note-trained patterns using several representations. Finally, Arzt and Widmer [2015] proposed a multi-agent-based online tracker using information from previously aligned performances of the same piece. On the second stage, online audio-to-score alignment is performed in two steps. First, the matching measure between events in the score and in the performance is defined. In particular, a cost matrix is estimated using a non-iterative signal decomposition method, previously developed by the authors in Carabias-Orti et al. [2013], that uses the spectral patterns fixed from the previous stage. Finally, we define an alignment strategy using online DTW and the speed of interpretation, which can be seen as a proxy for tempo changes. In particular, three different methods are proposed using this strategy: (1) BACKspeed approach that reports the onset times once they are detected (resulting in negative prediction times), (2) FWDspeed approach that reports the onset times with information of past note events (with positive prediction times), and (3) Accomp approach that estimates tempo changes from information of estimated note events.

These approaches have been submitted to the Music Information Retrieval Evaluation Exchange (MIREX) 2015 Realtime Audio-to-Score Alignment task and the results compared to previous editions, obtaining the best results in terms of total precision and piecewise precision. Moreover, a study of the method performance in function of the polyphony has also been performed to demonstrate the robustness of the proposed system.

The structure of the rest of the article is as follows. In Section 2, we briefly review the score following background with special attention to DTW-based approaches. In Sections 3 and 4, preprocessing and alignment stages of the proposed score following framework are explained, respectively. In Section 5, the evaluation set-up is presented, and in Section 6, the proposed method has been tested and compared with other reference systems. Finally, we summarize the work and discuss future perspectives in Section 7.

2. DTW FOR SCORE FOLLOWING BACKGROUND

Early works in score following were performed mainly using string-matching techniques [Dannenberg 1984]. Later, with the advent of faster computers, different approaches were developed using techniques such as DTW [Orio and Schwarz 2001; Hu et al. 2009; Dixon 2005], HMM [Cont 2006; Duan and Pardo 2011], hybrid graphical models [Raphael 2006], neural networks [Schreck-Ensemble 2001], or a conditional random field model [Joder et al. 2011].

In this article, we will use a low complexity signal decomposition method that obtains a distortion matrix for each score unit per frame. From this matrix, we directly compute the cost matrix used by a DTW algorithm to obtain the optimum path, and thus, perform the alignment. Consequently, a brief review of DTW is presented below.

DTW is a technique for aligning time series or sequences which has been intensively applied by the speech recognition community [Itakura 1975; Rabiner et al. 1978] and used in many fields such as data mining [Berndt and Clifford 1994] and information retrieval [Dannenberg and Hu 2003]. The series are represented by two vectors of features, $U = u_1, \dots, u_\tau, \dots, u_{T_m}$ and $V = v_1, \dots, v_t, \dots, v_{T_r}$, where τ and t are the point indexes in the time series. T_m and T_r represent the length of time series U and V , respectively. As a dynamic programming technique, it divides the problem into several sub-problems, each of which contribute in calculating the distance (or cost function) cumulatively.

The first stage in the DTW algorithm is to fill a local distance matrix (a.k.a. cost matrix) $\mathbf{D}(\tau, t) = \psi(u_\tau, v_t)$, where matrix $\mathbf{D}$ has $T_m \times T_r$ elements which represent the match cost between every two points in the time series. The cost function ψ could be any cost function that returns cost 0 for a perfect match, and a positive value otherwise (e.g., Euclidean distance).

In the second stage (forward step), a warping matrix $\mathbf{C}$ is filled recursively as:

$$\mathbf{C}(\tau, t) = \min \left\{ \begin{array}{c} \mathbf{C}(\tau - 1, t - 1) + \mathbf{D}(\tau, t) \\ \mathbf{C}(\tau - 2, t - 1) + \sigma_{2,1} \mathbf{D}(\tau, t) \\ \vdots \\ \mathbf{C}(\tau - \alpha_\tau, t - 1) + \sigma_{\alpha_\tau,1} \mathbf{D}(\tau, t) \\ \mathbf{C}(\tau - 1, t - 2) + \sigma_{1,2} \mathbf{D}(\tau, t) \\ \vdots \\ \mathbf{C}(\tau - 1, t - \alpha_t) + \sigma_{1,\alpha_t} \mathbf{D}(\tau, t) \end{array} \right\}, \quad (1)$$

where the step size at each dimension has a range from 1 to α_τ and 1 to α_t , respectively. α_τ and α_t are the maximum step size at each dimension. Parameter σ controls the bias toward diagonal steps as $\sigma_{x,y} = \sqrt{x^2 + y^2}$. $\mathbf{C}(\tau, t)$ is the accumulated cost of the minimum cost path from $(1, 1)$ to (τ, t) , and $\mathbf{C}(1, 1) = \mathbf{D}(1, 1)$.

Finally, in the last stage (traceback step), the minimum cost path $\mathbf{w} = w_1, \dots, w_L, \dots, w_L$ is obtained by tracing the recursion backwards from $\mathbf{C}(T_m, T_r)$. Each w_l is an ordered pair (τ_l, t_l) , such that $w_l \in \mathbf{w}$ means that the points u_{τ_l} and v_{t_l} are aligned. Moreover, the path has to satisfy the following three conditions:

- (i) Boundary condition: $w_1 = (1, 1)$ and $w_L = (T_m, T_r)$.
- (ii) Monotonicity condition: $\tau_{l+1} \geq \tau_l$ and $t_{l+1} \geq t_l$.
- (iii) Step size condition: $\tau_{l+1} \leq \tau_l + \alpha_\tau$ and $t_{l+1} \leq t_l + \alpha_t$.

The original formulation of Equation (1) considers $\sigma = 1$, so the path is biased towards diagonal steps with respect to vertical or horizontal steps. Some global path constraints have been introduced to avoid this behavior. For example, Sakoe and Chiba

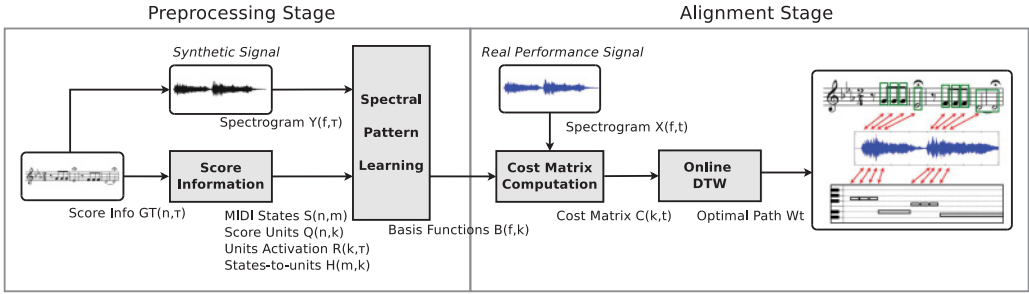


Fig. 1. Block diagram of the proposed score following framework. Images obtained from https://en.wikipedia.org/wiki/Music_alignment.

[1978] use a weight of $\sigma = 2$ for diagonal steps. Also, Itakura [1975] proposed to constrain the path to lie within a parallelogram around the diagonal of the matrix.

This approach has the advantage of being computationally simple and can be applied to audio-to-score synchronization. One of the first works that applied DTW to music alignment was presented in Orio and Schwarz [2001] following the standard definition of DTW and using Peak Structure Distance (PSD) as features. Another possibility is to use discrete chromagrams of the audio signal as features for both sequences, as shown in Dannenberg and Hu [2003]. Some approaches were proposed to reduce the complexity in time and space, such as Mueller et al. [2006] and Kaprykowsky and Rodet [2006]. Adaptive approaches have also been proposed to overcome structure changes by allowing partial synchronization path searches in the DTW alignment [Fremerey et al. 2010] or in the online context, by running several trackers in parallel [Arzt and Widmer 2010]. Finally, Dixon [2005] proposes an online DTW algorithm to follow piano performances, where each audio frame is represented by an 84 dimension vector, corresponding to the half-wave rectified first-order difference of 84 spectral bands. This onset-informed low-level feature works well for piano performances; however, for instruments with smooth onsets like string and wind, it may have difficulties.

3. PREPROCESSING STAGE

The proposed framework for realtime audio-to-score alignment is presented in Figure 1. As can be seen, the framework has two stages. First, the preprocessing stage must be computed beforehand and only the MIDI score is required. Then, once the parameters are learned, alignment can be computed in realtime. In this section, we explain the preprocessing stage of Figure 1, while the alignment stage will be explained in Section 4. In the preprocessing stage, the parameters for the alignment are learned from the score, which must be provided beforehand using MIDI representation. This stage is performed in two successive steps: score unit definition and spectral patterns learning, as detailed below.

3.1. Score Unit Definition

The aim of this step is to adequately organize the information given by the score to be used for alignment purposes.

First of all, the binary ground-truth transcription matrix $\mathbf{GT}(n, \tau)$ (see Figure 2(a)) is inferred from the MIDI score, where τ is the time in frames referenced to the score (MIDI time) and n are the notes in MIDI scale. In Figure 2(a), the MIDI score involves just one instrument (a piano), but more instruments can be defined in a score. For those cases, the n index refers to each note of the different instruments. Consequently, the number of total notes for a composition, N , is obtained as the sum over the different

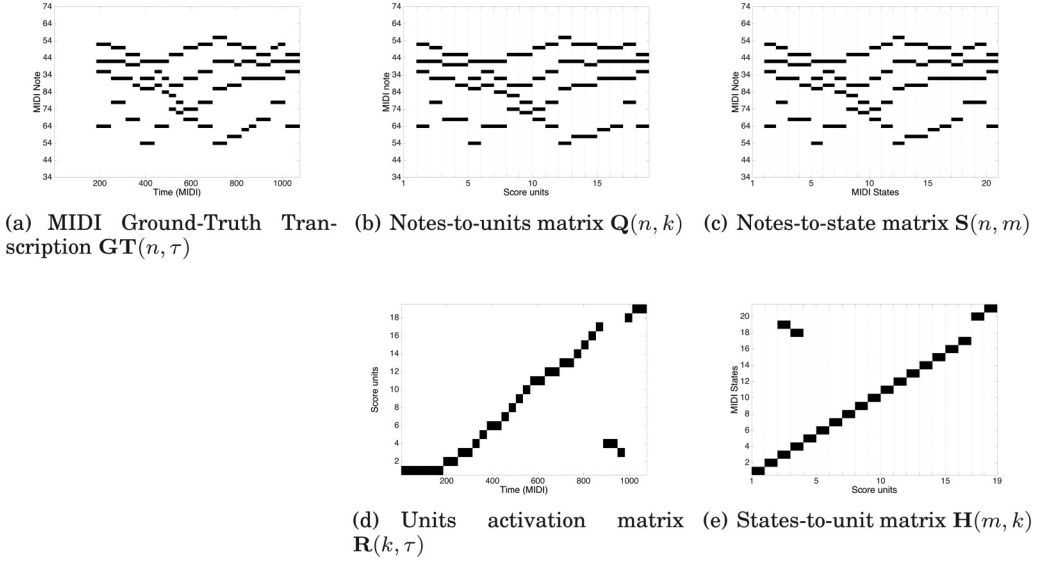


Fig. 2. Music signal from the test database in Section 4.1 (“01-AchGottundHerr”). (a) MIDI Ground-Truth Transcription $\mathbf{GT}(n, \tau)$. (b) Notes-to-units matrix $\mathbf{Q}(n, k)$. (c) Notes-to-states matrix $\mathbf{S}(n, m)$. (d) Units activation matrix $\mathbf{R}(k, \tau)$. (e) States-to-units matrix $\mathbf{H}(m, k)$.

notes per instrument. Each unique occurrence of individual or concurrent notes will be denoted here as a score unit or just unit. K is the number of units in the score and will be most likely smaller than the number of notes ($K \ll N$). The score defines a consecutive sequence of M states, where each state m is defined by a single unit. Note that opposite to the units, the states are not unique (i.e., multiple states can be represented by the same unit), and therefore, $K \leq M$. Also, the score informs about the time changes from one state to the next state. In fact, a score follower must determine the time (referenced to the input signal) of all transitions between states.

From the ground-truth transcription matrix $\mathbf{GT}(n, \tau)$, we obtain the following decomposition of binary matrices

$$\mathbf{GT}(n, \tau) = \mathbf{Q}(n, k)\mathbf{R}(k, \tau), \quad (2)$$

where $\mathbf{Q}(n, k)$ is the notes-to-units matrix, k is the index of each unique unit, and $\mathbf{R}(k, \tau)$ represents the activation of each unit in MIDI time. In Figure 2(b), the note-to-unit matrix $\mathbf{Q}(n, k)$ is represented. This matrix contains the notes belonging to each unit but no information about MIDI time. Conversely, $\mathbf{R}(k, \tau)$ matrix retains the MIDI time activation per unit but no information about the active notes per unit, as can be seen in Figure 2(d).

In order to obtain the information for MIDI states required to represent the sequence of units, the notes-to-units matrix $\mathbf{Q}(n, k)$ is further decomposed as

$$\mathbf{Q}(n, k) = \mathbf{S}(n, m)\mathbf{H}(m, k), \quad (3)$$

where $\mathbf{S}(n, m)$ is the notes-to-states matrix, m is the index for the MIDI states, M is the number of states, and $\mathbf{H}(m, k)$ represents the unique unit k active at each state m . In Figure 2(c), the notes-to-states matrix $\mathbf{S}(n, m)$ is represented. This matrix contains the notes belonging to each state, while $\mathbf{H}(m, k)$ matrix informs about the units active at each state, as can be seen in Figure 2(e).

The matrices here defined will be used in the next stages to perform the alignment and are computed from the MIDI score (see Section 4.2).

3.2. Spectral Patterns Learning

When a signal frame is given to a score follower, the first step should be the computation of a similarity measure between the current frame and the different units defined by the score. Our approach is to compute a distortion between the frequency transform of the input and just one spectral pattern per unit. A spectral pattern is here defined as a fixed spectrum which is learned from a signal with certain characteristics. The use of only one spectral pattern per unit allows us to compute the distortions with a low complexity signal decomposition method, though this assumption is not sufficient for modeling complex non-stationary sounds (e.g., the attack of a piano or a vibrato in a violin). This means that our method must learn in advance the spectral pattern associated to each unit of notes for the score. To this end, a state-of-the-art supervised method based on NMF with Beta-divergence and Multiplicative Update (MU) rules [Carabias-Orti et al. 2013] is used, but in this work, we propose to apply it on synthetic signal generated from the MIDI score instead of the real audio performance.

First of all, let us define the signal model as

$$\mathbf{Y}(f, \tau) \approx \hat{\mathbf{Y}}(f, \tau) = \mathbf{B}(f, k)\mathbf{G}(k, \tau), \quad (4)$$

where $\mathbf{Y}(f, \tau)$ is the magnitude spectrogram of the synthetic signal, $\hat{\mathbf{Y}}(f, \tau)$ is the estimated spectrogram, $\mathbf{G}(k, \tau)$ matrix represents the gain of the spectral pattern for unit k at frame τ , and $\mathbf{B}(f, k)$ matrix, for $k = 1, \dots, K$, represents the spectral patterns for all the units defined in the score.

When the parameters are restricted to be non-negative, as it is the case of magnitude spectra, a common way to compute the factorization is to minimize the reconstruction error between the observed spectrogram and the modeled one. The most popular cost functions are the Euclidean (EUC) distance, the generalized Kullback-Leibler (KL), and the Itakura-Saito (IS) divergences. Besides, the Beta-divergence (see Equation 5) is another commonly used cost function that includes in its definition the three previously mentioned EUC ($\beta = 2$), KL ($\beta = 1$), and IS ($\beta = 0$) cost functions.

$$D_\beta(x|\hat{x}) = \begin{cases} \frac{1}{\beta(\beta-1)}(x^\beta + (\beta-1)\hat{x}^\beta - \beta x\hat{x}^{\beta-1}) & \beta \in (0, 1) \cup (1, 2] \\ x \log \frac{x}{\hat{x}} - x + \hat{x} & \beta = 1 \\ \frac{x}{\hat{x}} + \log \frac{x}{\hat{x}} - 1 & \beta = 0 \end{cases} \quad (5)$$

In order to obtain the model parameters that minimize the cost function, Lee and Seung [2000] propose an iterative algorithm based on MU rules. Under these rules, $D_\beta(\mathbf{Y}(f, \tau)|\hat{\mathbf{Y}}(f, \tau))$ is shown to be non-increasing at each iteration while ensuring non-negativity of the bases and the gains. Details are omitted to keep the presentation compact. For further information, please read Lee and Seung [2000] and Févotte and Idier [2011].

Finally, the method to learn the spectral patterns for each state is described in Algorithm 1. Note that operator $\odot$ indicates Hadamard product (or element-wise multiplication), division and power are also element-wise operators, and $(\cdot)^T$ denotes matrix transposition.

As explained in Section 3.1, $\mathbf{R}(k, \tau)$ is a binary unit/time matrix that represents the activation of unit k at frame τ in the score. Therefore, at each frame, the active unit k is set to one and the rest are zero. Gains initialized to zero will remain zero, and therefore, the frame becomes represented with the correct unit.

An example of the process is displayed in Figure A.1. In particular, five different units can be found in the analyzed excerpt Figure A.1(a). Gains are initialized so only one unit is active at a time, and thus, sparsity is enforced in Figure A.1(b). From the synthetic signal, spectrogram is computed using semitone resolution time-frequency

ALGORITHM 1: Method for Learning Spectral Pattern Per Unit

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- 1: Initialize $\mathbf{G}(k, \tau)$ as the unit's activation matrix $\mathbf{R}(k, \tau)$ and $\mathbf{B}(f, k)$ with random positive values.
 - 2: Update the bases as $\mathbf{B}(f, k) \leftarrow \mathbf{B}(f, k) \odot \frac{(\mathbf{Y}(f, \tau) \odot \hat{\mathbf{Y}}^{\beta-2}(f, \tau)) \mathbf{G}^T(\tau, k)}{(\hat{\mathbf{Y}}^{\beta-1}(f, \tau)) \mathbf{G}^T(\tau, k)}$.
 - 3: Update the gains as $\mathbf{G}(k, \tau) \leftarrow \mathbf{G}(k, \tau) \odot \frac{\mathbf{B}(f, k) (\mathbf{Y}(f, \tau) \odot \hat{\mathbf{Y}}^{\beta-2}(f, \tau))}{\mathbf{B}(f, k) (\hat{\mathbf{Y}}^{\beta-1}(f, \tau))}$.
 - 4: Normalize each spectral pattern of $\mathbf{B}(f, k)$ to the unit β -norm.
 - 5: Repeat step 2 until the algorithm converges (or maximum number of iterations is reached).
-

representation and shown in Figure A.1(c). Finally, spectral pattern for the first unit is displayed in Figure A.1(d).

The case of the unit in which there are not any active notes must be initialized beforehand. In those MIDI frames where no signal is defined (silence frames) and in order to avoid numerical problems, we propose to initialize the spectral patterns associated to silence units to a constant. Note that a similar solution has been used in Gemmeke et al. [2011] to overcome the silence problem on an automatic speech recognition framework.

4. REALTIME ALIGNMENT STAGE

In this stage, the alignment between the score and the audio performance is accomplished in realtime using the information from the preprocessing stage.

4.1. Observation Model

As explained in Section 3.2, the spectral patterns $\mathbf{B}(f, k)$ for the K different units are learned in advance using a MIDI synthesizer and kept fixed. Each spectral pattern models the spectrum of a unique unit.

Now, the aim is to estimate the gain matrix $\mathbf{G}(k, t)$ and the cost matrix $\mathbf{D}(\tau, t)$ that measures the suitability of each unit to be active at each frame t (referenced to the signal input) by analyzing the likelihood between the spectral patterns $\mathbf{B}(f, k)$ and the input signal spectrogram¹. A similar approach was used in Fritsch and Plumbley [2013], but they use one component per instrument and note, plus some extra-components to model the residual sounds. Here, we use one component per unit so that single, non-zero restriction could be imposed to the gains, allowing the use of a very efficient signal decomposition method, as described below. From the cost matrix $\mathbf{D}(\tau, t)$, a classical DTW approach can be applied to compute the alignment path.

In this work, we propose to perform the factorization using the realtime single-pitch constrained method proposed in Carabias-Orti et al. [2013]. Although this method was designed to address music transcription of monophonic signals, it can be adapted for score following of polyphonic signals because only one unit will be active at a time. In this transcription method, the optimum component k_{opt} is chosen to minimize the Beta-divergence function at frame t under the assumption that only one gain is non-zero at each frame. Taking the unit as the index of gains $\mathbf{G}(k, t)$, this assumption is fair because only a single unit can be active at each time (at least when producing the audio signal).

Due to space limitation, equations to derive cost matrix $\mathbf{D}(\tau, t)$ using the approach in Carabias-Orti et al. [2013] are detailed in Appendix A.1. The process is summarized in Algorithm 2. Finally, we propose to use the distortion matrix $\mathbf{D}(\tau, t)$ as the input of the DTW algorithm in order to perform the alignment.

¹Note that we are using $\mathbf{X}$ and t instead of $\mathbf{Y}$ and τ to represent the signal magnitude spectrogram and the time frames to distinguish between real world and synthetic signals.

ALGORITHM 2: Distortion Matrix Computation Method

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1: Initialize  $\mathbf{B}(f, k)$  with the values learned in Section 3.2.
2: for  $t = 1$  to  $T$  do
3:   for  $k = 1$  to  $K$  do
4:     Compute the gains  $g_{k,t}$  using Equation (18).
5:     Compute the distortion values for the current unit in matrix  $\Phi(k, t)$  using Equation (19).
6:   end for
7: end for
8: Compute the cost matrix  $\mathbf{D}(\tau, t)$  between MIDI time and input signal time using
   Equation (20).

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4.2. Baseline DTW Path Computation

The classical formulation of DTW for score following requires two different inputs: the score and the audio signal from the performance. Here, we apply a common practice which is to treat the MIDI score as an audio signal by using synthesis [Dannenberg and Hu 2003].

We propose to use a DTW-based method to perform the alignment using the cost matrix $\mathbf{D}(\tau, t)$ obtained in Equation (20). This cost matrix is computed from the input signal $\mathbf{X}(f, t)$ and the “synthetic” spectral patterns per unit $\mathbf{B}(f, k)$, introduced in Section 3.2. The term “synthetic” comes from the fact that the spectral patterns $\mathbf{B}(f, k)$ are computed from the score using a MIDI synthesizer.

An example of the classical DTW performance is displayed in Figure A.2 (white dashed line). Details about the implementation of this method are given in Section 6.1 (referred to as Offline method). The y-axis represents the frame number of the synthesized MIDI signal (MIDI time), τ , while the x-axis represents the frame number of the audio signal, t , and the black solid line represents the minimum cost path computed with classical DTW, explained in Section 2. Note that the conditions imposed by DTW (see Section 2) to compute the optimum path allows one to match the MIDI time for each state in sequence with its corresponding real performance time.

4.3. Online DTW Approach

As commented in Section 1, DTW can be performed in an offline or online manner. The offline case goes through the whole input signal and then, when all the paths have been evaluated, the decision is made at the last frame. Finally, the optimum path $\mathbf{w}$ is selected. In fact, $\mathbf{w}$ relates each real signal frame t_i and its corresponding synthetic signal frame τ_i . Consequently, the alignment decision for every state cannot be known until the whole piece is processed. The online algorithm differs from a standard DTW algorithm in some points. First, the signal is partially unknown (i.e., signal values at future frames are not known when making the alignment decisions). Consequently, the global path constraints cannot be directly implemented; in other words, the recursion backwards can not be traced from the last pair (T_m, T_r) of synthetic and real signal frames. Second, the algorithm has a limitation in its anti-causal response. An online alignment system should be designed for limiting the maximum delay in reporting notes (i.e., a threshold in its negative prediction time values).

Here, we propose an online algorithm with a fixed latency of just one frame. In order to obtain this low latency, no backtracking is allowed; that is, the decision is made directly from the information at each frame t . Consequently, the obtained results with the proposed online approach are degraded from its offline counterpart. In fact, for those situations in which a higher latency is allowed, the use of a limited traceback could improve the alignment results at the expense of a complexity increment. In this

work, we focus on online alignment algorithms with positive prediction times, and therefore, traceback computation is above the scope of this article.

As explained in Carabias-Orti et al. [2015], the simplest online approach is obtained by matching the performance position at frame t with the score position τ associated to the minimum value of the accumulated cost $\mathbf{C}(\tau, t)$ at frame t (see Equation (1)). However, when different units coincide in most of their notes, as usually happens in polyphonic Western music, their corresponding spectral patterns will be highly correlated. In this case, similar distortion values are presented for the synthetic signal frames associated to those units in the cost matrix $\mathbf{D}(\tau, t)$. This situation is specially undesirable when similar units are activated in consecutive states at the score because it provokes a lack of precision for the DTW over these almost equally valued areas. Therefore, we must impose some constraints to compute an improved DTW path.

4.4. Online DTW with Anchor Points

In this work, we propose to select meaningful changes, in the same line as in Ewert et al. [2011], between states associated to units composed by different notes with low harmonic relation. We assume that the distortion values in the cost matrix $\mathbf{D}(\tau, t)$ should be considerably dissimilar between these states, so that the path estimation can be considered as reliable. Therefore, we propose to analyze each state onset time from the synthetic signal. We only consider a state onset time as reliable if the correlation value between the spectral pattern of its associate unit and the previous state one is below a certain threshold. These states' onset times are here denoted as anchor points. In order to determine the anchor points, we first obtain the spectral pattern for all the states in the score $\mathbf{BS}(f, m)$ as

$$\mathbf{BS}(f, m) = \mathbf{B}(f, k)\mathbf{H}^T(k, m), \quad (6)$$

where the states-to-units matrix $\mathbf{H}(m, k)$ represents the unit k active at each state m (see Section 3.2). The spectral pattern at m -th state can be represented as $\mathbf{bs}_m(f)$. Then, the normalized cross correlation between the spectral patterns associated to the current and the previous states, m -th and $m - 1$ -th, respectively, can be computed as

$$\mathbf{cc}(m) = \sum_f \mathbf{bs}_m(f) \cdot \mathbf{bs}_{m-1}(f). \quad (7)$$

Finally, we select as anchor points the onset MIDI times of those states that fulfill with the condition $\mathbf{cc}(m) < Th$. Consequently, the DTW path in the online case is estimated using these reliable positions (anchor points) which are represented as $\mathbf{w}^o = \{w_1^o, \dots, w_I^o\}$, where each w_i^o is a pair (τ_i, t_i) . In this way, the online DTW path $\mathbf{w}^o$ contains I pairs, where I is the number of anchor points. Each pair (τ_i, t_i) is estimated by assigning the MIDI time τ_i to the first signal frame t_i , at which the position of the minimum value of the accumulated cost $\mathbf{c}_i(\tau)$ is higher or equal than τ_i . Figure A.2 shows how anchor points are selected ($Th = 0.7$) and how they represent meaningful (τ_i, t_i) points along the DTW output path.

Anchor points are used as reference for tracking the speed of interpretation. Figure A.2 shows how the use of anchor points avoids online-DTW path mistakes over uncertainty areas of the cost matrix. Using the proposed approach, only a subset of score states are considered reliable and, consequently, an estimated value of the speed of interpretation is employed to assign onset times to those states between anchor points. A new value of speed of interpretation is computed when an anchor point is reached by the online-DTW path at frame t_i . The average speed value between anchor points i and $i - 1$, v_i^{interp} , is computed as

$$v_i^{interp} = (\tau_i - \tau_{i-1}) / (t_i - t_{i-1}). \quad (8)$$

This average speed value v_i^{interp} is used to update the speed of interpretation estimation at anchor point i by following Equation (9).

$$v_i^{estim} = \alpha_v \cdot v_i^{interp} + (1 - \alpha_v) \cdot v_{i-1}^{estim} \quad (9)$$

The speed of interpretation estimation is a commitment value between change adaptation and filtering spurious values of average speed. v_i^{interp} values which are out of the 95% interval confidence are discarded from the v_i^{estim} computation. $\alpha_v = 0.1$ has demonstrated to give a good balance. The initial value for the speed of interpretation estimation is set to 1.

4.5. Algorithm Variants

All the proposed methods follow the same strategy until an anchor point i is reached and the new speed of interpretation is updated. Then, the way this information is used allows for three different variants for the score following decisions.

—**BACKspeed:** Once the speed of interpretation is updated at a new anchor point, the real onset times of those states whose score onset times are between t_{i-1} and t_i are estimated in the BACKspeed variant. This means that t_i is the decision time so the prediction time is always negative (i.e., the real onset times of states between anchor points $i - 1$ and i are reported after t_i is reached). The output for real onset time is then computed as:

$$t^{onset} = ((\tau^{onset} - \tau_{i-1})/v_i^{estim}) + t_{i-1}, \quad (10)$$

where t^{onset} is the real onset time assigned to the state, τ^{onset} is the midi onset time of this state, and v_i^{estim} is the speed of interpretation estimation for the i -th anchor point.

—**FWDspeed:** This variant is introduced to inform about onset times between anchor points with positive prediction times. Consequently, the real onset times of states between anchor points i and $i + 1$ are reported by following Equation (11).

$$t^{onset} = ((\tau^{onset} - \tau_i)/v_i^{estim}) + t_i \quad (11)$$

Now, the onset times for states between anchor points i and $i + 1$ are reported when t_i is reached, resulting in positive prediction times (i.e., onset times are predicted before their estimated times). Figure A.7 at Appendix A.3 shows the difference between BACKspeed and FWDspeed based on how the speed of interpretation is applied to compute the states' onset times.

—**Accomp:** The two previous versions report onset times as output messages. Alternatively, Accomp method is designed for reporting tempo messages as outputs. In this way, a synthesizer can be used as an instrumental accompaniment of a human musician. Consequently, this approach does not estimate states' onset times between anchor points; on the contrary, it just establishes the speed of interpretation at each anchor point. We have followed the next four conditions in the Accomp variant design:

- (i) The prediction times must be positives.
- (ii) The synthesizer tempo can not be zero.
- (iii) The synthesizer tempo can not tend to infinity.
- (iv) The synthesizer tempo messages are only sent at anchor point times.

The first condition implies to use the anchor point information as in the FWDspeed variant. Therefore, the synthesizer tempo of states between anchor points i and $i + 1$ are reported when t_i is reached. The second condition implies that we can not stop the synthesizer and wait until the DTW path reaches the next anchor point, even though

the synthesizer reaches the midi onset time of the next anchor point τ_{i+1}^{synth} before the DTW reports t_{i+1} . The third condition implies that we can not set the synthesizer at time t_{i+1} , though the DTW reports t_{i+1} before the synthesizer reaches the midi onset time of the next anchor point τ_{i+1}^{synth} . The fourth condition is introduced for limiting the tempo messages to coincide with anchor point times t_i . Here, we have denoted as τ_i^{synth} the MIDI time for the anchor point i at the synthesizer. In fact, τ_i^{synth} is aimed to be as close as possible to the MIDI time of the anchor point τ_i at frame t_i .

The first parameter to be computed in the Accom variant once the DTW informs about the pair (τ_i, t_i) is the MIDI time of the synthesizer at frame t_i as

$$\tau_i^{synth} = (t_i - t_{i-1})v_{i-1}^{synth} + \tau_{i-1}^{synth}, \quad (12)$$

where τ_{i-1}^{synth} is the MIDI time of the synthesizer at frame t_{i-1} and v_{i-1}^{synth} is the tempo of the synthesizer between frames t_{i-1} and t_i . Also, the prediction for the onset time of the next anchor point state t_{i+1}^{predic} is computed as

$$t_{i+1}^{predic} = ((\tau_{i+1} - \tau_i)/v_i^{estim}) + t_i, \quad (13)$$

where the same approximation as Equation (11) is used. This prediction time will not match perfectly to (τ_{i+1}, t_{i+1}) . Because of that, when the anchor point $i + 1$ is reached (at time t_{i+1}), the synthesizer timer will be at a different MIDI time τ_{i+1}^{synth} .

Finally, the tempo message to be sent to the synthesizer is computed following Equation (14).

$$v_i^{synth} = (\tau_{i+1} - \tau_i^{synth})/(t_{i+1}^{predic} - t_i), \quad (14)$$

where the tempo v_i^{synth} is set in order to have a synthesizer MIDI time τ_{i+1} at frame t_{i+1}^{predic} . In case the tempo v_{i-1}^{synth} brings the system to overtake the following anchor point in MIDI time at frame t_i (i.e., $\tau_i^{synth} > \tau_{i+1}$), the tempo in Equation (14) would be computed by ignoring the overtaken anchor points; in other words, τ_{i+1} would be changed by the MIDI time of the next anchor point after τ_i^{synth} .

This process is repeated for all anchor points. Figure A.5 at Appendix A.3 shows how the score following system sends messages of tempo changes to a synthesizer which is supposed to always process notes with the last informed tempo value. Figure A.3 at Appendix A.3 shows an example of the comparison between the ground-truth tempo and the Accom tempo which would be sent to the synthesizer. It can be seen that the system supports the performance tempo changes.

The Accom variant is expected to have a worse performance than the other ones because instead of onset times, it is driven by tempo messages. However, it is the most realistic approach for automatic music accompaniment. Finally, an analysis of the limitations of the proposed framework is explained in Appendix A.6

5. EVALUATION

5.1. Datasets

To evaluate the proposed methods we use the MIREX² realtime audio-to-score alignment database. This dataset has not changed for campaigns from 2010 to 2014, allowing a fair comparison between the submitted algorithms. Concretely, the evaluation is performed using three datasets composed by recordings of human played performances

²The Music Information Retrieval Evaluation eXchange (MIREX) is an annual evaluation campaign for MIR algorithms. Real-time Audio-to-Score Alignment (a.k.a. Score Following) is one of the evaluation tasks. <http://www.music-ir.org/mirex>.

together with their corresponding MIDI representations of the score. Recordings are in 44.1KHz 16 bit wav format.

- (1) The first one is composed of 46 recordings extracted from four distinct musical pieces: vocals from W.A. Mozart opera pieces, J.S. Bach Sonatas and Partitas for solo violin, W.A. Mozart Oboe Quartet in F major, and flute from P. Boulez ...explosante-fixe....
- (2) The second one consists of three piano performances of the Prelude in G minor op. 23 – 5 by Sergei Rachmaninoff.
- (3) The third one is composed of 10 human played J.S. Bach four-part chorales. The audio files are sampled from real music performances recorded at 44.1kHz that are 30 seconds in length per file. Each piece is performed by a quartet of instruments: violin, clarinet, tenor saxophone, and bassoon. Each musician's part was recorded in isolation. Individual lines were then mixed to create 10 performances with four-part polyphony. More information about this dataset can be found in Duan and Pardo [2011].

Here, the third subset is used to optimize parameter β (see Equation (5)) and analyze the performance of the proposed methods against polyphony. The Bach10 dataset has been kindly shared by the authors of Duan and Pardo [2011]. Note that the pieces from the other two datasets are not public to prevent overtraining. From the Bach Chorales dataset, individual lines were mixed to create audio files containing all duets (60 files), trios (40 files), and four-part polyphony (10 files), totaling 110 test files. The ground-truth is obtained using a robust single pitch detection algorithm YIN [de Cheveign and Kawahara 2002] for each individual part. Then, some manual corrections were held afterwards to fix some apparent errors.

5.2. Experimental Setup

- (a) Time-Frequency representation: In this article, we use a low-level spectral representation of the audio data which is generated from a windowed Fast Fourier Transform (FFT) of the signal. A Hanning window with the size of 128ms, and a hop size of 10ms is used (for both synthetic and real-world signals). Here, we use the resolution of a single semitone. In particular, we implement the time-frequency representation by integrating the STFT bins corresponding to the same semitone.
- (b) Evaluation metrics: We have used the same evaluation metrics as in the MIREX Real-time Audio-to-Score Alignment task. Detailed information can be found in Cont et al. [2007]. For each piece, Aligned Rate (AR) or precision is defined as the proportion of correctly aligned notes in the score and ranges from 0 to 1. A note is said to be correctly aligned if its onset does not deviate more than a threshold (a.k.a. tolerance window) from the reference alignment. Typical threshold values range from 100 to 300ms. Missed notes are events that are present in the reference but not reported. Recognized notes whose onsets are far from the given threshold are considered misaligned notes.

Other metrics characterizing the quality of an alignment are:

- Miss rate: percentage of missed score events.
- Misalign rate: percentage of misaligned events.
- Average offset: average sign-valued time offset between a reported note onset by the score follower and its real onset in the reference file.
- Average abs offset: average absolute-valued time offset.
- Std offset: standard deviation of sign-valued time offset.
- Average prediction time: average difference between the onset time assigned to a note event by the alignment system and the time it reports about this note event.

The measures above are calculated both globally (over the whole database) and locally (for each sound file). Thus, two precision rates are considered:

—Overall precision rate: percentage of correctly aligned score events on the whole database.

—Piecewise precision rate: average of the percentage of detected notes for each piece.

- (c) MIDI synthetic signals generation: In this work, two kinds of MIDI data can be used for the preprocessing stage, using specific instrument information or not. In fact, the General MIDI (GM) standard allows to define a concrete instrument for each event and provides a standardized sound bank that allows a Standard MIDI File created on one device to sound similar when played back on another. In the case that no instrument is defined in the MIDI file, the piano instrument is used by default. The effects of using specific instrument information or default piano for score alignment are analyzed in Appendix A.4.3. Then, MIDI synthetic signals are generated using Timidity++ with the FluidR3 GM soundfont on Mac OS.

5.3. Implementation

The DTW stage of Section 2 has been implemented following a *client-server* architecture, as described in Alonso et al. [2016]. It uses a round buffer which stores only α_r columns of the accumulated cost matrix $\mathbf{C}(\tau, t)$ for the online case. This implementation has been tested over x64/x86 architecture devices, ARM[®] architecture devices, and GPUs. Tests have been run with a score file which represents some hours of music performance (Goldberg Variations). Then, results show the maximum score length that each device can compute in order to accomplish a maximum latency of 10ms. Some descriptive examples are here remarked. An Intel[®] Xeon[®] E5-2603 device supports up to two hours (7,200sec) of performances with only 2 cores. A NVIDIA Shield Tablet (ARM[®] Cortex[®] A15 architecture with 4 cores) supports up to 5 minute performances using only one core, whereas it supports 15 minute performances when using 3 and 4 cores. In the case of GPUs, results do not outperform results obtained with the other tested architecture. A whole analysis of the implementation, performance, and computational cost is explained in Alonso et al. [2016].

6. RESULTS

6.1. Results on J.S. Bach Chorales Database

First of all, we analyzed the effect of the different parameters for the proposed methods in Section 4.5. To this end, we carried out several tests using the third subset of the MIREX database, detailed in Section 5.1. Evaluation has been performed using the metrics detailed in Section 5.2. Here, a 300ms tolerance window has been applied, as proposed by Cont et al. [2007]. Due to space limitation, the study of parameter optimization is detailed in the Appendix A.4. In fact, optimal performance has been found for $\beta = 0$ and $Th = 0.7$.

Now, we aim to evaluate the performance of the proposed methods in function of the polyphony. To this end, the proposed systems (FWDspeed, BACKspeed, and Accomp) are compared with four reference methods that are detailed below:

- (1) Offline: This method represents the offline alignment using DTW and was first introduced in Carabias-Orti et al. [2015]. The $\tau \times t$ cost matrix $\mathbf{D}(\tau, t)$ is estimated using Equation (20). The optimum path is obtained by tracing the recursion backwards from $\mathbf{C}(T_m, T_r)$, as in the original formulation of DTW (see Section 2). For the DTW algorithm, we have used the code from Ellis [2003].
- (2) ISMIRon [Carabias-Orti et al. 2015]: This method represents the online alignment using DTW without introducing anchor points. This online approach is obtained by

Table I. Audio-to-Score Results as a Function of Polyphony in Terms of Piecewise Precision

	Poly	Precision	Miss	Misalign	Av. Offset	Av. Abs Offset	Std Offset	Av. Pred. Time
FWDspeed	2	84,18	0,00	15,81	-42,83	70,83	76,72	201,12
	4	85,66	0,00	14,44	-47,47	70,07	75,01	210,34
	3	86,70	0,00	13,30	-43,11	69,79	75,64	223,82
BACKspeed	2	88,11	0,00	11,89	-28,79	60,58	69,43	-219,31
	3	89,22	0,00	10,77	-27,13	59,99	70,75	-216,31
	4	91,59	0,00	8,40	-32,88	63,61	71,73	-222,07
Accomp	2	69,75	0,00	30,25	-84,43	127,94	114,32	514,73
	3	69,57	0,00	30,42	-91,90	137,62	118,86	516,07
	4	69,13	0,00	30,86	-92,82	140,42	121,28	542,07
ISMIRon	2	88,93	0,00	11,07	65,80	84,76	84,76	0
	3	91,28	0,00	8,72	66,35	85,76	86,02	0
	4	91,70	0,00	8,30	64,74	85,28	84,33	0
Soundprism	2	85,36	0,00	14,64	-27,93	53,35	65,50	-
	3	89,63	0,00	10,37	-17,09	45,19	57,87	-
	4	94,65	0,00	5,35	-23,31	45,29	54,75	-
Offline	2	95,85	0,00	4,15	-6,51	35,15	53,23	-
	3	95,78	0,00	4,22	-10,47	36,53	51,35	-
	4	96,12	0,00	3,88	-9,67	38,92	55,88	-
Oracle	2	100,00	0,00	0,00	8,98	30,43	37,51	-
	3	100,00	0,00	0,00	10,99	32,17	40,49	-
	4	100,00	0,00	0,00	6,01	32,60	43,41	-

The bold percentage shows the best result for each measure.

matching the performance position at frame t with the score position τ associated to the minimum value of the accumulated cost $C(\tau, t)$ at frame t . This matching is done at each frame t to obtain the DTW path in the online case, as explained in Carabias-Orti et al. [2015].

- (3) Soundprism [Duan and Pardo 2011]: Although this method is designed for sound source separation, it includes an audio-to-score alignment stage which is evaluated here for comparison. This system uses a hidden Markov process to model the audio performance. The state space is defined as a 2D space of score position and tempo. The observation model is defined as the multi-pitch likelihood of each frame, i.e., the likelihood of seeing the audio frame given the pitches at the aligned score position. Particle filtering is employed to infer the score position and tempo of each audio frame in an online fashion.
- (4) Oracle system: From the third dataset of Section 5.1, the oracle system is obtained by interpolating the MIDI beat times and the annotated times by tapping ones from the audio. The latter were estimated manually by a musician. Note that, besides the tempo differences between audio and MIDI, the audio often includes a fermata after each musical phrase. Therefore, the oracle system assumes perfect synchronization and the same tempo between the musicians while the MIDI has a constant tempo. This system acts as a “global ground-truth,” and thus, represents the best performance that can be obtained by any score follower. Here, this oracle system is very useful to analyze the deviation of the different instrument performances between themselves.

Numerical results are displayed in Table I. As could be seen, the Offline method obtains the best results, although the latency coincides with the duration of the signal, and thus, it is not an alternative for real-time/low latency applications.

Regarding the online methods, BACKspeed, ISMIRon, and Soundprism obtain similar results on average and clearly outperform FWDspeed and Accomp, although our

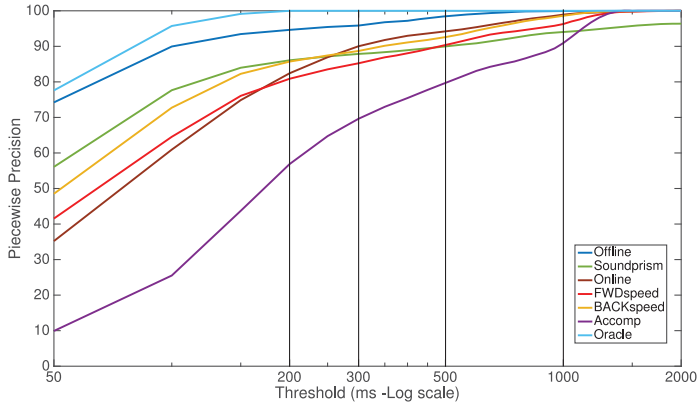


Fig. 3. Precision values in function of the onset deviation threshold.

approaches seem to be more robust against the polyphony. Accom results are quite degraded in comparison to FWDspeed. ISMIRon, which uses an online version of DTW algorithm without anchor points, obtains slightly better results than BACKspeed. For the Bach Chorales dataset, the optimum value of Th parameter (see Appendix A.4) is about 0.9 and the proposed threshold ($Th = 0.7$) obtains slightly worse results than not using anchor points ($Th = 1$). Unlike the proposed methods, Soundprism can not be implemented in real-time, as stated in Duan and Pardo [2011], due to the computational complexity derived from the particle filtering algorithm. Moreover, Soundprism obtained very high precision values in the MIREX Score Following task for the analyzed dataset, although performance drops considerably for other pieces, as demonstrated in Figure 9(b) (detailed info can be seen in the MIREX 2010 results webpage).

It is very interesting to see that the miss rate for the studied methods is 0. In fact, our proposed methods and Soundprism uses tempo information, and therefore, all the notes in the score will be reported in the same order indicated in the score.

In relation to the offset, all values expressed in Table I are in ms. The Oracle solution exhibits the minimum possible std offset due to the differences in starting times for the same MIDI states between musicians. Moreover, Accom obtains the highest std offset. In fact, enforcing the path to avoid disruptive tempo changes causes our method to be less responsive, and therefore, the results fall drastically when evaluating with lower threshold values.

Finally, we have evaluated the precision of the analyzed methods as a function of the onset deviation threshold. To this end, the threshold value was varied from 50 to 2,000ms in 50ms steps. Obtained results are plotted in Figure 3. As can be seen, the Oracle method, which can be considered as the best performance that we can obtain with any Score Follower, requires around a 200ms threshold to obtain a perfect alignment. This value comes from the difference between the ground-truth alignment for each instrument played in isolation and the global ground-truth of the whole mixture, obtained by interpolating the annotated beat times of each audio. In general, the Offline method performs almost as good as the Oracle, clearly outperforming the online approaches. Within the online approaches, Soundprism seems to perform better than the other online approaches when using low threshold values. Then, up to 200ms BACKspeed and ISMIRon become the most accurate online approaches. Unlike BACKspeed, FWDspeed and Accom provide positive prediction times, i.e., report the event onset times before they actually occur, and therefore, obtain slightly worse results. Finally, the Accom method is specifically designed for automatic accompaniment, and therefore, its results

Table II. Simulation Configuration

Campaign	Algorithms/Metric	Overall Precision	Piecewise Precision
2015	RVC6 (BACKspeed)	95.70	92.99
	RVC5 (FWDspeed)	95.53	92.92
	RVC4 (Accomp)	92.99	84.24
2014	RV3	90.98	93.55
	RV2	89.28	90.51
	CR1	82.17	84.35
2013	<i>CVRCR1</i>	86.70	82.09
	CJ5	67.10	78.30
2012	CC1	67.10	78.30
	YSK1	52.81	46.38
	<i>CRVCC1</i>	83.01	77.44
2011	JC1	64.90	68.54
	SUROS1	67.11	63.83
2010	AW1	50.84	50.33
	DP1 (Soundprism)	49.11	67.14
	RVCC3	32.17	62.79
	RVCC4	32.44	64.50
	SUROS1	73.97	73.93

The bold percentage shows the best result for each measure.

are considerably degraded when dealing with pieces with sudden tempo changes. In fact, these kind of pieces are usually employed to stress the alignment methods.

6.2. MIREX Real-Time Audio-to-Score Alignment Results

Table II provides the results from last year (2010-2015) in the MIREX Real-time Audio-to-Score Alignment (a.k.a. Score Following) task³.

For this task, we submitted the three proposed online algorithms to the 2015 campaign. Moreover, FWDspeed, BACKspeed, Accomp, and the state-of-the-art Soundprism [Duan and Pardo 2010] are denoted in MIREX as “RVC4,” “RVC5,” “RVC6,” and “DP1.” Moreover, approaches “RV2” [Rodriguez-Serrano and Vera-Candeas 2014] and “RV3” [Rodriguez-Serrano and Vera-Candeas 2014] submitted by the authors in MIREX 2014 can be seen as previous versions of “FWDspeed” and “BACKspeed,” respectively.

The other online⁴ systems presented in the MIREX 2010-2015 campaigns are “SUROS1” [Suzuki et al. 2010], “RVCC3” [Rodriguez-Serrano et al. 2010], “RVCC4” [Rodriguez-Serrano et al. 2010], “AW1” [Arzt and Widmer 2010], “SUROS1” [Suzuki et al. 2011], “JC1” [Chen and Jang 2011], “YSK1” [Yamamoto et al. 2012], “CC1” [Chen 2012], “CC1” [Chen 2013], and “CR1” [Romani and Carabias-Orti 2014].

As can be seen, the proposed BACKspeed and FWDspeed systems obtain the best results in terms of Overall and Piecewise Precision. The Accomp system also provides competitive results. Note that parameters in the 2015 versions of FWDspeed and BACKspeed are optimized, with respect to the previous year ones in order to better estimate the tempo changes. Although in some pieces sudden variations could produce a slight under-performance of the current versions, in general, relying on the tempo estimation provides better results by means of overall precision. Extended MIREX results are given at Appendix A.5.

³Early results for MIREX 2006 could be found in http://www.music-ir.org/mirex/wiki/2006:Score_Following_Results, but evaluation was not performed using the full database proposed in 2010.

⁴Offline approaches marked with * in MIREX are not considered in the study.

7. CONCLUSIONS

In this article, we present a score following framework based on spectral factorization and online DTW. The proposed framework is based on two stages: preprocessing and alignment. In the first stage, the MIDI score is analyzed in order to define the different combinations of concurrently played notes (a.k.a. score units) and the transitions between them (i.e., the sequence of states). Then, an NMF-based method with Beta-divergence is applied to learn spectral templates for all units from a synthetic MIDI signal. In the second stage, a cost matrix is computed using the distortion matrix obtained using a non-iterative signal decomposition method, previously developed by the authors in Carabias-Orti et al. [2013], that has been tuned to perform the projection of each unit. Finally, the alignment is obtained using a strategy based on online DTW, where the optimal path is biased by the speed of interpretation.

In order to deal with different applications, three methods are presented using this framework, BACKspeed, FWDspeed, and Accomp. BACKspeed is intended for applications that allow a certain delay between the occurrence of the note event and the time it is reported by the system. FWDspeed and Accomp are intended for positive prediction time applications, such as automatic accompaniment; in other words, the prediction is made before the actual occurrence of the note event using the information about the speed of interpretation inferred from the previous note event times.

To study the limitations of both approaches, we tested them using a dataset with different polyphony levels (from two to four) and compared them with other reference methods. On the one hand, BACKspeed provides accurate results when a negative prediction time is allowed. On the other hand, Accomp method provides a suitable alternative to deal with automatic accompaniment, where a positive prediction time is required.

Finally, to test the reliability of the proposed methods, we submitted them to the MIREX 2015 Realtime Audio-to-Score Alignment task. Among the compared methods, BACKspeed obtained the best results in terms of overall precision and piecewise precision. These results can be extrapolated to campaigns from 2010 to date, since the evaluation is performed using the same datasets.

In the future, we plan to improve Accomp method by relaxing the design conditions.

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